UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

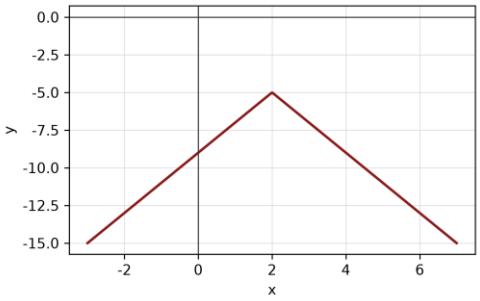
INSTRUCTIONS TO MARKERS

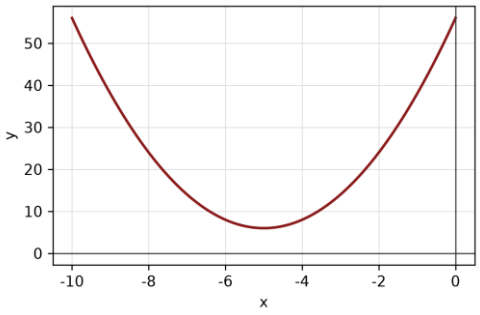
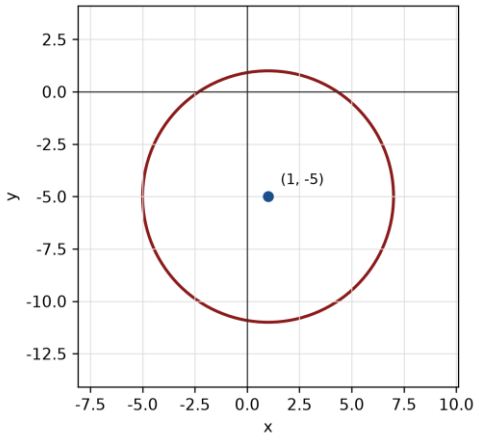
1. This marking scheme accompanies the MAT133 mock paper (October 2026).
2. Answers and marking points are for local study use only.

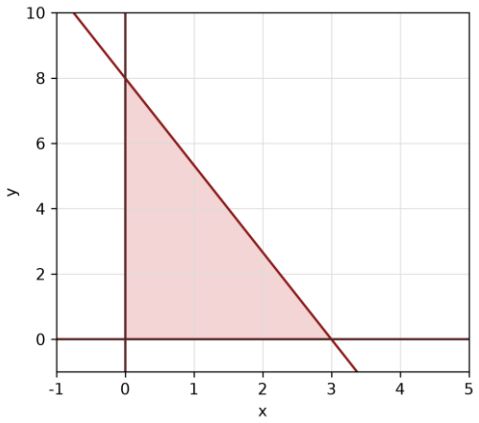
FOR MARKERS' USE ONLY

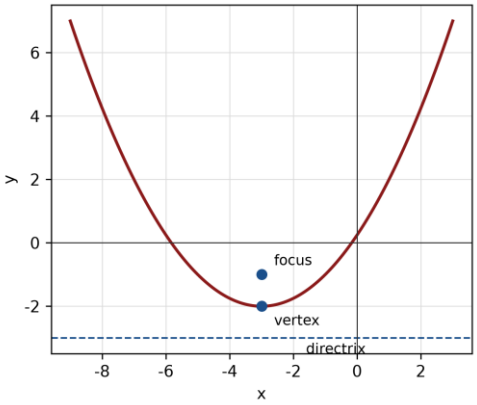
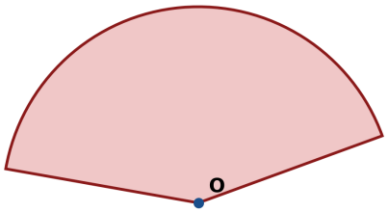
This marking scheme consists of 6 printed pages

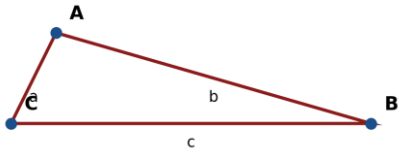
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NO	SOLUTION	MARK
1 a)	$-3x-4<-4$ M1 $-3x<0$ $(0,\infty)$ A1 	3 m
1 b)	$\lvert 4x-5\rvert-4<0$ M1 $-4<4x-5<4$ A1 $\left(\frac{1}{4},\frac{9}{4}\right)$ A1 	5 m
1 c)	$y=-2\lvert x-2\rvert-5$ B1 vertex $(2,-5)$ A1 no real intercepts A1 	5 m

NO	SOLUTION		MARK
2 a)	$y = 2(x+5)^2 + 6$ vertex $(-5, 6)$ $y(0) = 56$ 	B1 B1 A1 A1	5 m
2 b)	$f(x) = x^5 - 3x^3 - 6x$ $f(-x) = -x^5 + 3x^3 + 6x$ $-f(x) = -x^5 + 3x^3 + 6x$ $f(-x) = -f(x)$	M1 A1 A1	4 m
2 c)	$x^2 - 2x + y^2 + 10y = 10$ $(x-1)^2 + (y+5)^2 - 26 = 10$ $(x-1)^2 + (y+5)^2 = 36$ 	M1 M1 A1 A1	5 m

NO	SOLUTION		MARK
3 a)	$\frac{-3+3i}{1-i}$ $= \frac{(-3+3i)(1+i)}{(1-i)(1+i)}$ $= \frac{-6}{2}$ -3	M1 M1 A1 A1	5 m
3 b)	$\lvert z\rvert = \sqrt{8^2+15^2} = 17$ $\tan\theta = \frac{15}{8}, \quad \theta = 61.9^\circ$	M1 A1 M1 A1	5 m
3 c)	$-3x^3-5x^2+x+3 = (x+4)(-3x^2+7x-27)+111$ $q(x) = -3x^2+7x-27, \quad r = 111$	M1 M1 A1 A1	5 m
4 a)	$\frac{x}{3}+\frac{y}{8}=1$ $x \geq 0, \quad y \geq 0, \quad 8x+3y \leq 24$ 	M1 A1 A1	5 m
4 b)	$-6x-12 = x^2-4$ $x^2+6x+8=0$ $(x+2)(x+4)=0$ $x=-4$ or $x=-2$ $(-4,12), \quad (-2,0)$	M1 A1 A1	5 m

NO	SOLUTION		MARK
4 c)	$y = \frac{x^2}{4} + \frac{3x}{2} + \frac{1}{4}$ $y = \frac{(x+3)^2}{4} - 2$ $(x+3)^2 = 4y + 8$ 	M1 A1 A1 A1	5 m
5 a)	$\cos A = -\frac{12}{13}, \quad \sin B = -\frac{5}{13}$ $\sin 2A = 2\sin A\cos A = -\frac{120}{169}$ $\cos(A+B) = \cos A\cos B - \sin A\sin B = -\frac{119}{169}$	M1 M1 A1 A1	5 m
5 b)	$\theta = 150^\circ \times \frac{\pi}{180} = \frac{5\pi}{6}$ $s = r\theta = 13\left(\frac{5\pi}{6}\right) = \frac{65\pi}{6}$ 	M1 M1 A1	4 m

NO	SOLUTION	MARK
5 c)	$\frac{a}{\sin A} = \frac{b}{\sin B}$ **M1** $\frac{13}{\sin 45^\circ} = \frac{b}{\sin 60^\circ}$ **M1** $b = \frac{13\sin 60^\circ}{\sin 45^\circ} = \frac{13\sqrt{6}}{2}$ **A1** 	**4 m**